DEEP LEARNING FOR EEG MOTOR IMAGERY: A SYSTEMATIC COMPARISON WITH SUBJECT-WISE EVALUATION

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ABSTRACT

Brain-computer interfaces, or BCIs, are based on electroencephalography motor imaging. They offer a non-invasive pathway for assistive access and neural rehabilitation. This work presents a systematic comparative study of four deep learning architectures using Vanilla RNN, LSTM, CNN, and Transformers for three-class motor imagery classification using the PhysioNet EEG Motor Movement/Imagery Dataset comprising of 109 subjects across six motor imagery runs. The isolated neural frequencies center around the Beta-band power features (13–30 Hz) and were extracted from electrodes C3, Cz, and C4 using the Welch periodogram, and a strict subject-wise train/validation/test split (70/15/15) was enforced to prevent data leakage, a methodological consideration frequently overlooked in the EEG deep learning literature. A hyperparameter search across 192 configurations was conducted from hidden size, number of layers, dropout rate, optimizer, learning rate, and architecture-specific parameters including kernel size and filter count for CNN and model dimension for Transformer. CNN achieved the highest mean test accuracy (41.23% $\pm$ 1.21%) and significantly outperformed all other architectures ($p < 0.0001$), demonstrating superior consistency across hyperparameter configurations. The Transformer achieved the highest single-configuration peak accuracy (43.21%) but exhibited the greatest variance (std $= 3.01\%$), indicating high sensitivity to hyperparameter selection. An ablation study also conducted confirmed that random level splitting artificially inflates test accuracy due to subject identity leakage. All models significantly exceeded the chance level performance of 33.3% ($p < 0.0001$), and per-class analysis showed difficulty distinguishing left from right hand imagery, motivating future work incorporating hemispheric asymmetry features and spatial filtering techniques.

1 INTRODUCTION

Brain-computer interfaces (BCIs) allow for direct communication between the brain and external devices, offering assistive and rehabilitative opportunities for individuals with severe motor impairments, including those who've undergone strokes and neurological deficits, for example (Wolpaw et al., 2002). Among non-invasive BCI paradigms, motor imagery (MI) based on electroencephalography (EEG) is good because it does not require actual movement and relies on naturally occurring sensorimotor rhythms (Pfurtscheller & Neuper, 2001). During MI, event-related desynchronization (ERD) and synchronization (ERS) in the beta band (13–30 Hz) can be observed over contralateral sensorimotor areas (Pfurtscheller & Lopes da Silva, 1999), providing a strong neural measure for classifying imagined movements. These brain areas undergo a decrease in power detection when a person either moves or imagines moving.

Deep learning has been increasingly applied to EEG decoding, with convolutional architectures demonstrating strong performance on motor imagery classification tasks (Schirrmeister et al., 2017; Lawhern et al., 2018). More recently, Transformer-based models have emerged as a strong direction for EEG analysis, allowing self-attention mechanisms to capture temporal dependencies (Song et al.,

2023). However, many studies unknowingly introduce data leakage by splitting samples randomly across subjects, leading to overly optimistic performance estimates. Moreover, a comparison of different architectures under identical, controlled conditions with proper subject-wise evaluation remains limited.

In this work, I present a subject-wise evaluation of four deep learning architectures, Vanilla RNN (Elman, 1990), LSTM (Hochreiter & Schmidhuber, 1997), CNN (LeCun et al., 1998), and Transformer (Vaswani et al., 2017), on a three-class motor imagery task using the publicly available PhysioNet dataset (Goldberger et al., 2000; Schalk et al., 2004). I extract beta-band power features from three electrodes (C3, Cz, C4) and enforce a strict subject-wise split to prevent leakage. Through a comprehensive hyperparameter search covering 192 configurations, I identify the best settings for each model, analyse stability, and examine per-class and per-subject performance. Some key contributions are: (1) leakage-free subject-wise evaluation techniques shown through ablation, (2) a systematic 192-configuration comparison of four architectures with statistical significance testing, and (3) a per-subject variability analysis to measure the challenge of subject-independent generalization.

2 METHODS

2.1 DATASET AND PREPROCESSING

For this evaluation, I use the EEG Motor Movement/Imagery Dataset (Goldberger et al., 2000; Schalk et al., 2004) from PhysioNet, containing recordings from 109 subjects. Six runs per subject were selected (runs 4, 6, 8, 10, 12, 14) corresponding to motor imagery trials. Three classes are defined: T0 (rest), T1 (left hand motor imagery), and T2 (right hand motor imagery). Raw EEG data were loaded using MNE, channel names standardized, and the standard 10-20 montage applied (Jasper, 1958). A bandpass filter of 13–30 Hz was applied to isolate beta-band activity (Pfurtscheller & Lopes da Silva, 1999), and epochs were extracted from -2.0 to $+2.0$ seconds relative to cue onset with baseline correction over -2.0 to -1.0 seconds. Given Mu has high variability, the primary focus centers around this Beta band to reduce subject variability in results across the 109 subjects. Only three electrodes were tracked, C3, Cz, and C4, which lie on top of the primary motor cortex and capture contralateral activation patterns during motor imagery (Pfurtscheller & Neuper, 1997; Jasper, 1958). Each epoch was truncated to 513 time samples. The full dataset yielded 19,010 epochs: T0 9,184, T1 4,939, T2 4,887.

2.2 FEATURE EXTRACTION

Raw EEG signals were downsampled by a factor of 2 from 160 Hz to 80 Hz, giving 256 time points per epoch. Beta-band power features were computed using a sliding window of 30 samples (0.375 seconds at 80 Hz) with a step size of 8 samples (73% overlap). For each window and each of the three channels, the power spectral density (PSD) was estimated using the Welch method with `nperseg` equal to the window size, and mean power in the 13–30 Hz range was computed. This produced a feature sequence of shape $(N, 29, 3)$ 29 temporal windows and 3 channels per epoch.

2.3 DATA SPLITTING AND NORMALIZATION

The class imbalance in T0 was corrected by undersampling to the mean of T1 and T2 counts, giving approximately 14,739 balanced epochs. The 109 subjects were partitioned into training (76 subjects, 70%), validation (16 subjects, 15%), and test (17 subjects, 15%) sets at the subject level, so no subject's data appears across partitions. Features were normalized using `StandardScaler` fitted exclusively on the training set. Final dataset sizes gave a train of 10,222 epochs, validation 2,198 epochs, test 2,319 epochs. Data loaders used batch size 64.

2.4 MODEL ARCHITECTURES

All models receive input of shape $(B, 29, 3)$ and output a 3-class logit vector. **Vanilla RNN** (Elman, 1990): A standard recurrent network with tanh nonlinearity. The hidden state sequence is mean-pooled across all time steps, followed by the Batch Normalization (Ioffe & Szegedy, 2015), Dropout

(Srivastava et al., 2014), and a linear classification head. **LSTM** (Hochreiter & Schmidhuber, 1997): Follows the same setuop as the RNN, replacing simple recurrent cells with LSTM units containing forget, input, and output gates to better capture temporal dependencies and reduce vanishing gradients. **CNN** (LeCun et al., 1998): Three 1D convolutional blocks, each applying a convolution of configurable kernel size $k \in \{3, 5, 7\}$ followed by Batch Normalization and ReLU. Filter counts double per block starting from base $f \in \{32, 64\}$. Adaptive average pooling reduces temporal dimension to 4, followed by two fully connected layers with Dropout. **Transformer** (Vaswani et al., 2017): A linear input projection maps the 3-channel input to a d-dimensional embedding, where sinusoidal positional encodings are added. Positional encoding is critical as self-attention is otherwise permutation-invariant for input order, making it blind to the temporal order. The encoder uses `norm_first=True` for training stability. Output is mean-pooled across time, followed by Dropout and a linear head.

2.5 Training Details

All models were trained with cross-entropy loss. I evaluated two optimizers: AdamW (Loshchilov & Hutter, 2019) with weight decay 10^{-2}, and SGD with momentum 0.9 and the same weight decay. A `ReduceLROnPlateau` scheduler halved the learning rate when validation accuracy plateaued for 5 epochs (minimum lr 10^{-6}). Gradient clipping at maximum norm 1.0 was used at every step. Early stopping used patience of 15 epochs during sweeping and 20 epochs during final retraining. Maximum epochs were 80 during sweeping and 100 during final retraining. All experiments were run on an NVIDIA P100 GPU in Kaggle.

3 Experiments

3.1 Hyperparameter Search

A grid search was performed across 192 total configurations, 48 per model:

- **RNN / LSTM**: hidden size $\in \{64, 128, 256\}$, layers $\in \{1, 2\}$, dropout $\in \{0.3, 0.5\}$, optimizer $\in \{\text{AdamW}, \text{SGD}\}$, learning rate $\in \{0.001, 0.0005\}$.
- **CNN**: kernel size $\in \{3, 5, 7\}$, base filters $\in \{32, 64\}$, dropout $\in \{0.3, 0.5\}$, optimizer $\in \{\text{AdamW}, \text{SGD}\}$, learning rate $\in \{0.001, 0.0005\}$.
- **Transformer**: $d_{\text{model}} \in \{64, 128\}$ with $n_{\text{head}} \in \{4, 8\}$ respectively, layers $\in \{2, 3\}$, dropout $\in \{0.3, 0.4\}$, optimizer $\in \{\text{AdamW}, \text{SGD}\}$, learning rate $\in \{0.001, 0.0005, 0.0001\}$. Three learning rate values were evaluated for the Transformer since its pretty sensitivite to learning rate selection (Vaswani et al., 2017).

The best configurations I found were: RNN (hidden=64, layers=1, dropout=0.3, AdamW, lr=0.0005), LSTM (hidden=256, layers=2, dropout=0.3, AdamW, lr=0.0005), CNN (kernel=3, filters=32, dropout=0.3, AdamW, lr=0.001), Transformer (d=64, heads=4, layers=2, dropout=0.3, AdamW, lr=0.001).

3.2 Evaluation Protocol

Test accuracy was computed on held-out test subjects using the best validation accuracy configuration per model. Paired t-tests and the Wilcoxon signed-rank tests evealuated statistical significance across all 48 configurations per model. One-sample t-tests against chance (33.3%) verified that each model learned a meaningful signal. Per-class accuracy was computed separately for T0, T1, and T2. A per-subject accuracy analysis was conducted on the 17 test subjects using the best CNN model. An ablation study compared subject-wise versus random sample-level splitting to quantify data leakage effects and maximise data.

Table 1: Model performance across 48 hyperparameter configurations each. Best Acc is the highest single-configuration test accuracy. Mean and Std are computed across all 48 configurations.

Model	Best Acc	Mean Acc	Std	Min Acc
Vanilla RNN	41.60%	37.87%	2.30%	33.46%
LSTM	42.08%	39.14%	1.78%	35.20%
CNN	43.00%	41.23%	1.21%	37.38%
Transformer	43.21%	38.38%	3.01%	32.90%
Chance	33.33%	—	—	—

4 Results

4.1 Overall Performance

Table 1 and Figure 1 summarize performance across all configurations. CNN achieved the highest mean test accuracy (41.23% $\pm$ 1.21%) and significantly outperformed all other architectures ($p < 0.0001$, both paired t-test and Wilcoxon signed-rank test). The Transformer outputted the highest single-configuration peak accuracy at (43.21%) but also vshowed the greatest variance with a (std $= 3.01\%$), highlighting a high sensitivity to the hyperparameter selection. LSTM greatly outperformed RNN (paired t-test $p = 0.0066$; Wilcoxon $p = 0.0411$). No statistically significant differences were found between Transformer and LSTM ($p = 0.1115$) or between Transformer and RNN ($p = 0.4049$). All models exceeded chance-level performance of 33.3% ($p < 0.0001$).

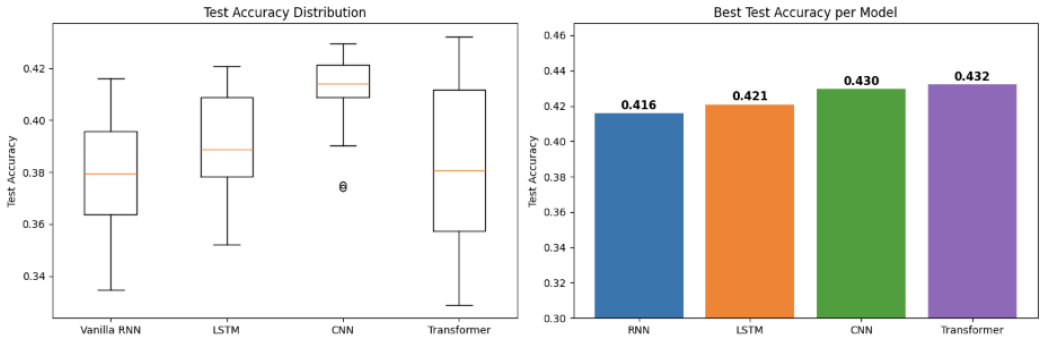

Figure 1: Model comparison. Left: test accuracy distribution across all 48 hyperparameter configurations per model. Right: best single-configuration test accuracy per model.

4.2 Learning Curves

Figure 2 shows the learning curves for the best configuration of each model. CNN exhibits mild overfitting, with training accuracy consistently exceeding validation accuracy, while RNN, LSTM, and Transformer show more stable generalization behavior.

4.3 Hyperparameter Analysis

For RNN and LSTM, AdamW consistently outperformed SGD across all configurations, while SGD performance approached chance level for several settings. For RNN, larger hidden sizes lowered performance, with hidden size 64 performing the best, while LSTM showed the opposite trend with performance increasing with hidden size up to 256. This difference suggests LSTM's gating mechanism better utilizes additional capacity while the vanilla RNN is greatly impacted from vanishing gradients at larger hidden sizes. Lower learning rates (0.0005) generally outperformed higher rates (0.001) for RNN, while LSTM showed resilience to learning rate choice. These trends are visualized in Figure 3.

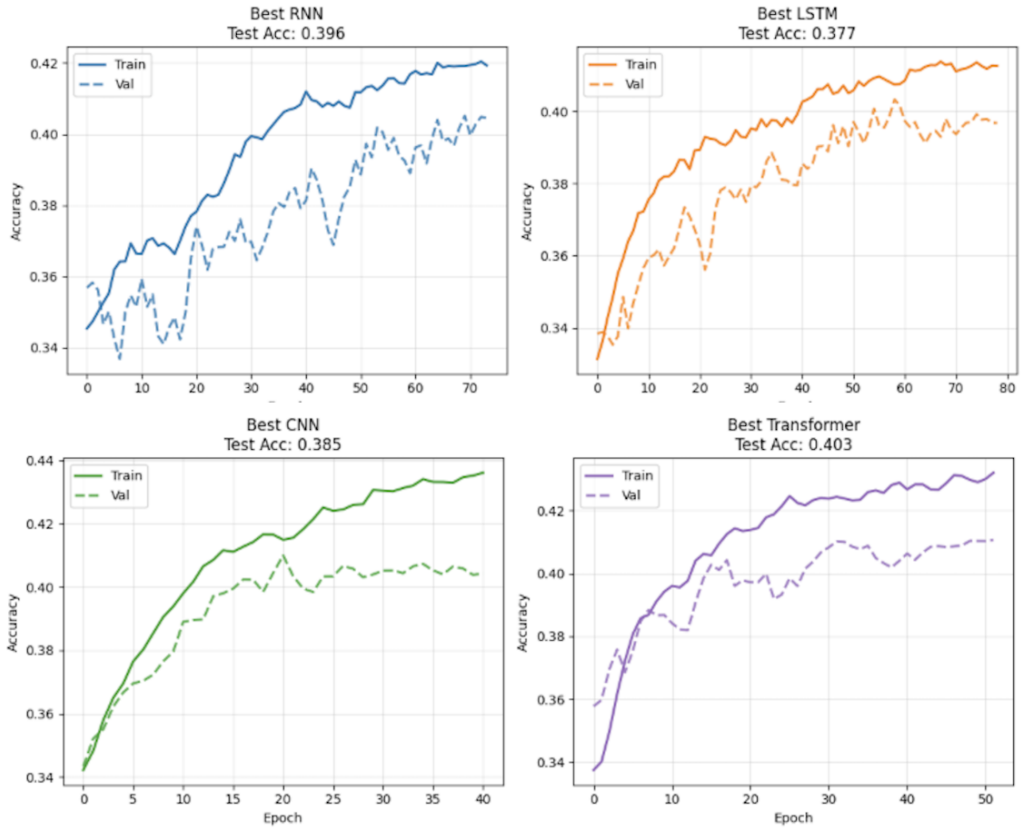

Figure 2: Learning curves for the best configuration of each model. Solid lines show training accuracy, dashed lines show validation accuracy.

For the Transformer model, SGD consistently collapsed to near-chance performance across all configurations, this confirms that adaptive optimization is essential for attention-based models on this task. The learning rate analysis revealed a non-monotonic relationship among Adam configurations, with $\text{lr} = 0.0001$ performing best while $\text{lr} = 0.0005$ yielded the lowest mean accuracy. So adding new inputs would not change the outputs but likley affect the findings. Model dims showed a small positive effect, with $d_{\text{model}} = 128$ slightly outdoing $d_{\text{model}} = 64$. Figure 4 shows the effect of each hyperparameter for the Transformer under Adam optimization.

For CNN, kernel size 5 achieved the highest mean accuracy among Adam configurations, outperforming both kernel size 3 and the larger kernel size 7, suggesting a modest temporal receptive field is optimal for 29-step beta bandpower sequences. Lower learning rates again performed better, consistent with findings across all models. Figure 5 showing the effect of kernel size and learning rate on CNN performance.

4.4 Per-Class Analysis

Figures 6 and 7 show confusion matrices and per-class accuracy across all models. Rest (T0) was the most reliably classified class (49–60%), while Right Hand (T2) was consistently the hardest (15–49%). The Transformer exhibited extreme class imbalance: 60% Rest accuracy and 49% Right Hand accuracy, but only 12% Left Hand accuracy. CNN demonstrated the most balanced performance: Rest 51%, Left Hand 47%, Right Hand 18%.

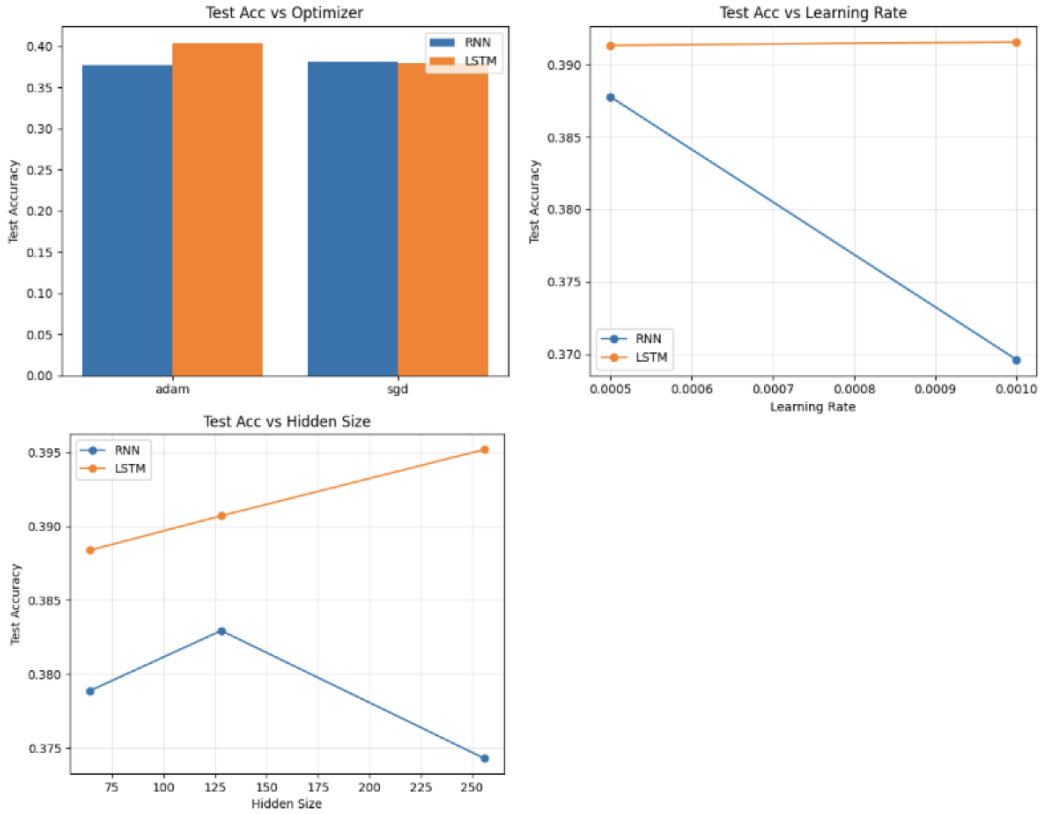

Figure 3: Hyperparameter analysis for RNN and LSTM. Effects of optimizer, learning rate, and hidden size on mean test accuracy.

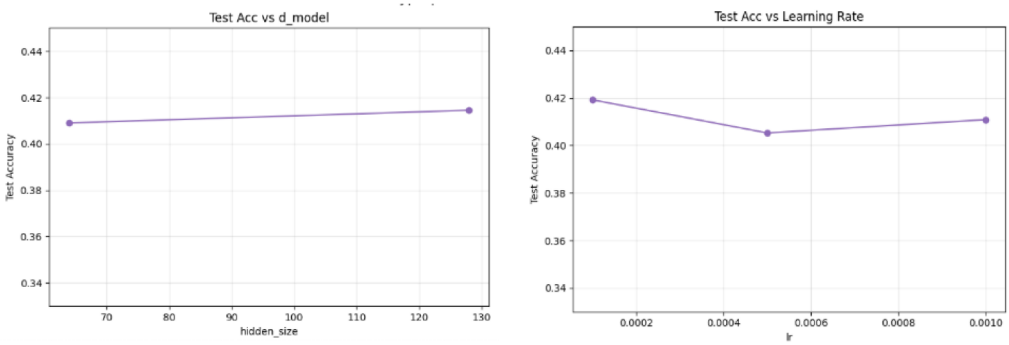

Figure 4: Transformer hyperparameter analysis (Adam optimizer only). Effects of model dimension and learning rate on mean test accuracy.

4.5 Per-Subject Variability

Per-subject accuracy using the best CNN model showed high individual variability among the 17 test subjects, ranging from 26.3% to 50.4% with a mean of 38.5%. Only 14 of 17 subjects exceeded chance level, meaning that 3 subjects were effectively unclassifiable by the subject-independent model.

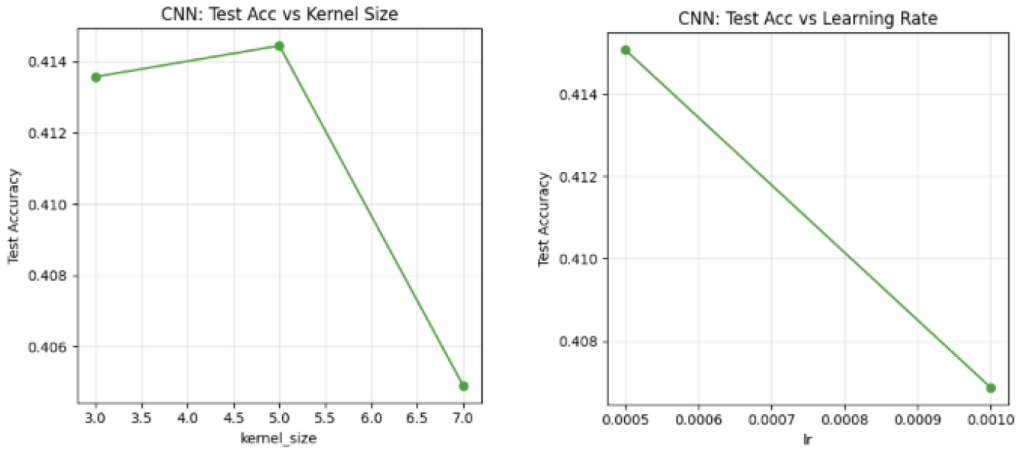

Figure 5: CNN hyperparameter analysis (Adam optimizer only). Effects of kernel size and learning rate on mean test accuracy.

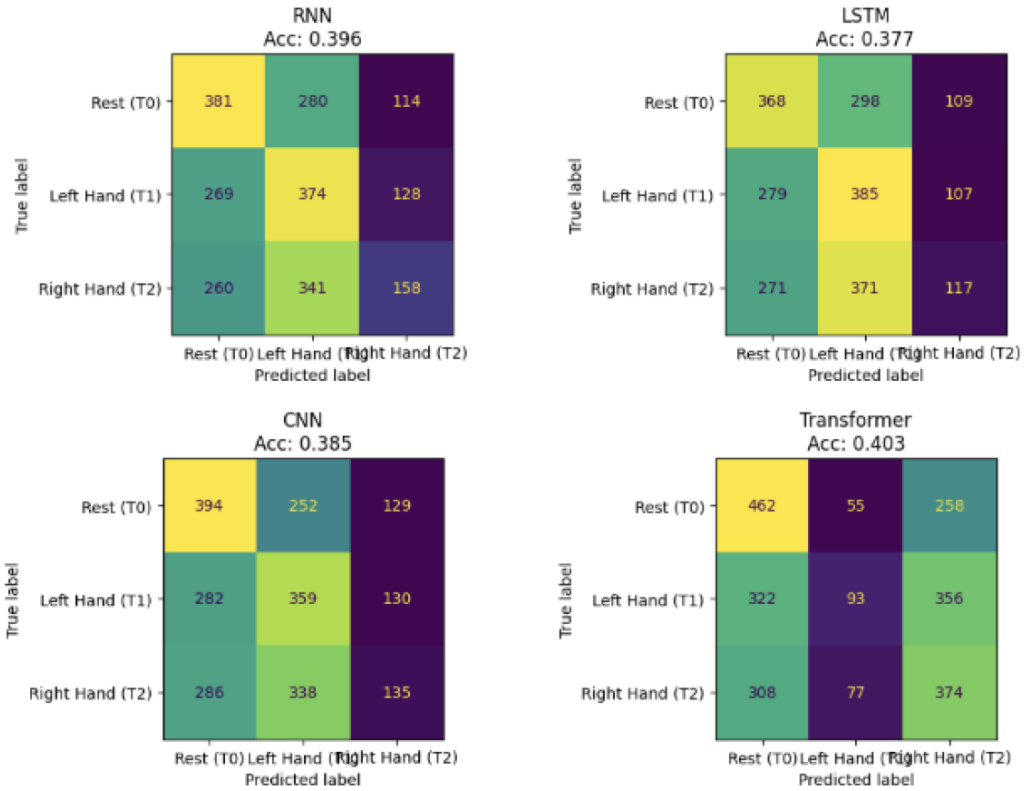

Figure 6: Confusion matrices for the best configuration of each model on the held-out test set.

4.6 Ablation Study

Retraining best configurations using normal random sample-level splitting confirmed that accuracy is artificially inflated due to subject identity leakage across all four models, underscoring the importance of subject-wise evaluation for honest generalization assessment in EEG-based BCI research.

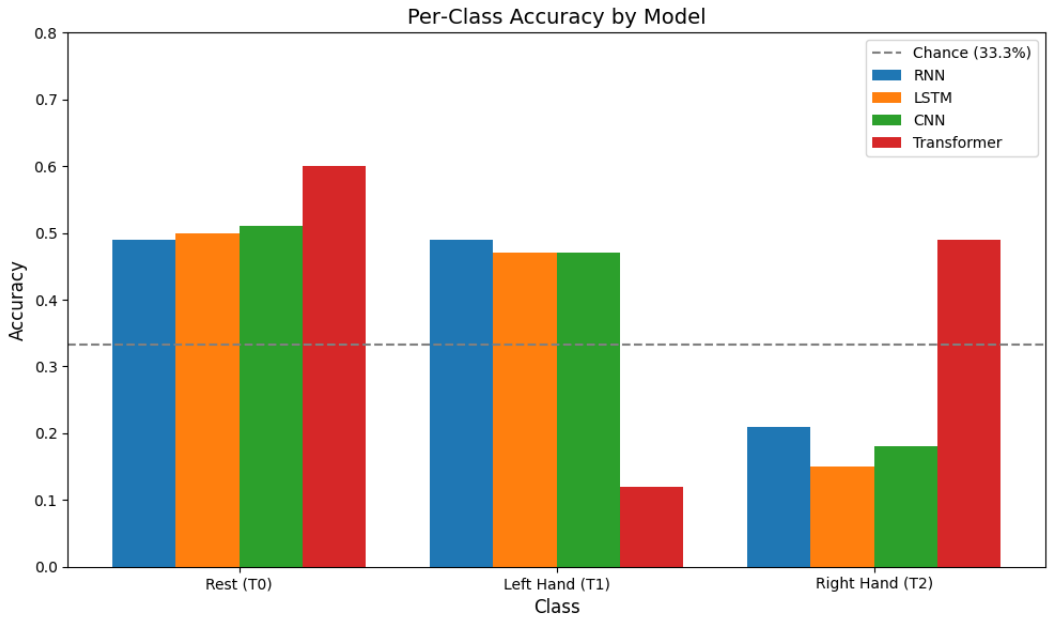

Figure 7: Per-class accuracy for Rest (T0), Left Hand (T1), and Right Hand (T2) across all four models. Dashed line indicates chance level (33.3%).

5 Conclusion

I present a comparative study of four deep learning architectures for three-class EEG motor imagery classification on the PhysioNet dataset comprising 109 subjects. Using a strict subject-wise evaluation method and a hyperparameter search across 192 configurations, CNN achieved the highest mean test accuracy (41.23%) and most consistent performance (std 1.21%), grealty outperforming all other architectures ($p < 0.0001$). The Transformer achieved the highest peak accuracy (43.21%) but exhibited the greatest sensitivity to hyperparameter choice (std 3.01%), with SGD collapsing to near-chance performance across all Transformer configurations. LSTM significantly outperformed Vanilla RNN ($p = 0.0066$), consistent with the theoretical advantage of gating mechanisms for sequential data. An ablation study confirmed that random splitting increases accuracy due to subject identity leakage, indicating a method concern relevant to the broader EEG deep learning literature. Per-class analysis revealed that right hand imagery classification remains consistently difficult across all architectures, motivating future work on hemispheric asymmetry features, Common Spatial Pattern filtering, and subject-specific adaptation.

Limitations

Several limitations of this study should be acknowledged.

First, the feature extraction pipeline relies only on beta-band power computed via the Welch periodogram across three electrodes (C3, Cz, C4). While this approach is computationally efficient and neuroscientifically motivated (Pfurtscheller & Lopes da Silva, 1999), it disregards temporal fine structure within each window and ignores inter-channel relationships such as hemispheric asymmetry. A laterality index defined as $(C4 - C3)/(C4 + C3)$ or Common Spatial Pattern filters could better visualize the contralateral activation patterns known to occur during left and right hand motor imagery (Pfurtscheller & Neuper, 1997), potentially improving classification of T1 versus T2 which all models struggled with across every experimental configuration.

Second, the per-class accuracy analysis revealed a consistent difficulty classifying Right Hand (T2) across all model architectures, with per-class accuracies for Right Hand ranging from 15–49% across architectures, compared to 49–60% for Rest (T0) and 12–49% for Left Hand (T1), indicating that right hand imagery was the most consistently difficult class to classify. The Transformer exhibited

particularly severe class imbalance 60% Rest accuracy but only 12% Left Hand accuracy, suggesting attention-based models may redistribute classification capacity poorly across classes, achieving 49% Right Hand accuracy while nearly failing on Left Hand entirely. This asymmetry shows that simple per-channel bandpower features are not enough to reliably distinguish lateralized motor imagery without explicit spatial filtering (Pfurtscheller & Neuper, 1997).

Third, the surface Laplacian spatial filter, which reduces volume conduction effects between neighboring electrodes, was not applied. With only three channels this would be an approximation, but could nonetheless improve spatial specificity of C3 and C4 recordings by reducing signal contamination from surrounding cortical areas, a common method when analyzing (EEG) data

Fourth, all models were trained in a subject-independent manner using a subject-wise split. While procedurally correct for evaluating generalization, the per-subject accuracy analysis revealed large individual variability among the 17 test subjects, ranging from 26.3% to 50.4% with a mean of 38.5%, and only 14 of 17 subjects exceeded chance level. This variability highlights that subject-specific fine-tuning or transfer learning approaches could greatly improve individual classification performance and show a natural direction for future work.

Finally, all models operate on pre-extracted bandpower features rather than raw EEG signals. Learning curve analysis revealed small overfitting in the CNN model, with training accuracy consistently exceeding validation accuracy, meaning that regularization techniques beyond dropout and weight decay may be warranted for larger feature sets. End-to-end deep learning approaches such as EEGNet (Lawhern et al., 2018) or ShallowConvNet (Schirrmeister et al., 2017), which learn spatial and temporal filters directly from raw signals, may achieve higher accuracy on this dataset and represent a greater approach for future investigation than the four models used in this analysis..

Bonus Points

I think that I deserve a few extra points given I utilize 192 configurations across four architectures (RNN, LSTM, CNN, Transformer) testing hidden size, layers, dropout, optimizer, learning rate, kernel size, filters, and model dim. I apply methods for subject-wise split with an ablation study proving random splitting increasing accuracy, this addresses a real problem in EEG research that not many consider, and because my per-subject analysis goes beyond just reporting averages, I analyzed individual variability across 17 test subjects, showing the challenge of subject-independent generalization (26.3–50.4Totally up to grader but I think this work adds to meaningful investigations for improvements in those who need it most!

Appendix

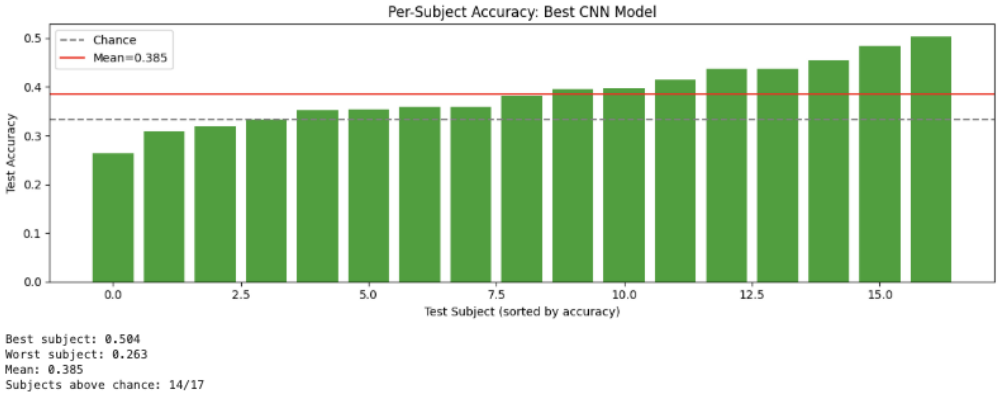

Figure 8: Per-subject test accuracy using the best CNN model across 17 held-out test subjects, sorted by accuracy. Red line indicates the mean accuracy (38.5%). Dashed line indicates chance level (33.3%). Only 14 of 17 subjects exceeded chance level, highlighting substantial inter-subject variability in subject-independent motor imagery classification.

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